

Denon AVR control protocol

Last Update: Jun. 30. 2015

Ver.02

Application terminal: Ethernet/RS-232C

Model	Terminal	Beacon No
AVR-X1200W	Ethernet	12.4.0
AVR-X2200W	Ethernet/	12.5.0
AVR-X3200W	Ethernet/RS-232C	12.6.0
AVR-X4200W	Ethernet/RS-232C	12.7.0

Connector specification

I. RS-232C

Connector type: DB-9pin female type, slave straight connection (DCE type)
(1pin : GND , 2pin : TxD , 3pin : RxD , 5pin : Common(GND) , 4,6,7,8,9pin : NC)

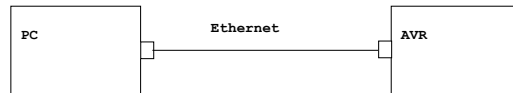
Communication format:

Synchronous system : Tone step synchronization
Communication system : A half duplex
Communication speed : 9600bps
Character length : 8 bits
Parity control : None
Start bit : 1 bit
Stop bit : 1 bit
Communication procedure : Non procedural
Communication data length : 135 bytes (maximum)

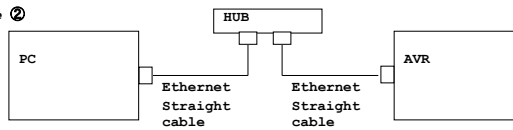
II. Ethernet

Connector type : RJ-45(10BASE-T/100BASE-TX)

Example ①



Example ②



Communication format :
Communication system : A half duplex
Communication speed : 10Mbps/100Mbps
Communication port : TCP port 23 (telnet)
Communication data length : 135bytes (maximum)

NETWORK SETUP of AV Receiver

>Procedure of Network Setup mode.

(1) Press **SETUP** button, then Menu appears on FL-display (and GUI)

(2) Select "Network > Settings > " .

(3) Set parameters described below.
<DHCP> "ON"---Use this setting when DHCP server is on the local network.

<IP Address> When <DHCP> sets "Off", please set IP address.

<Subnet Mask> When <DHCP> sets "Off", please set Subnet Mask.

<Gateway> Set the address of Gateway when Gateway is on the local network.

<Primary DNS> Do not set this parameter.

<Second DNS> Do not set this parameter.

<Proxy> Set this parameter "Off".

<Network Option: Network Standby Mode>

(1) Press **SETUP** button, then Menu appears on FL-display (and GUI)

(2) Select "Network > Network Control" (except for X6200)

Select "Network > IP Control" (X6200)

(3) Set this parameter "Always On".

"Always On"---Use this setting when using the AV Receiver Connected in a network.

Always respond to network commands.

"Off In Standby"--- Use this setting when not using the AV Receiver connected in a network.

Ignores network commands during standby to save power.

Protocol specification

The following three data forms are defined.

- COMMAND** : The message sent to a system(AVR) from a controller(Touch Panel etc.)
A command to a system is given from a controller.
Send the COMMAND in 50ms or more intervals.
- EVENT** : The message sent to a controller (Touch Panel etc.) from a system (AVR)
The result is sent, when a system is operated directly and a state changes.
The EVENT should be sent within 5 seconds after the state of the system (AVR) is changed.
*The form of EVENT presupposes that it is the same as that of COMMAND.
**Refer to the following table for the contents of COMMAND and EVENT.
- RESPONSE** : The message sent to a controller (Touch Panel etc.) from a system (AVR)
if the 'request command' (COMMAND+? +CR (0x0D)) has come from a controller.
The RESPONSE should be sent within 200ms of receiving the request COMMAND.
*The form of RESPONSE presupposes that it is the same as that of EVENT.

Basic specification: The command by ASCII CODE, parameter expression

*ASCII CODE which can be used is from 0x20 to 0x7F:
the alphabet and the number of 0-9, and space (0x20), some signs,
AND carriage return (0x0D) --- It is used only as a pause sign.

Command structure: COMMAND + PARAMETER + CR (0x0D)

COMMAND: ASCII CODE of 2 characters

Ex. SI : Select Input source
MS : surround Mode Setting
MV : Master Volume setting
PW : system Power setting

PARAMETER : ASCII CODE (up to 25 characters)

Ex. DVD : function name
STEREO : surround mode name
*Special Parameter--- ? : for request command

The example of a command * <CR> is the meaning of 0x0D.

SIDVD<CR> : Select Input source DVD
MSSTEREO<CR> : surround Mode Set to STEREO
MVUP<CR> : Master Volume UP
PWON<CR> : system Power ON
PWSTANDBY<CR> : system Power STANDBY
SI?<CR> : Request command for now playing input source >> Return RESPONSE 'SI***<CR>'

Others

- A) COMMAND is receivable also during transmission of EVENT.
- B) Since CHANNEL VOLUME changes simultaneously when the input source changes, the value of the channel volume of used channels returns as EVENT.
- C) Since SURROUND MODE or CHANNEL VOLUME changes simultaneously when the INPUT source changes, the SURROUND MODE or CHANNEL VOLUME returns as EVENT.
- D) When SURROUND MODE or CHANNEL VOLUME is the same in between INPUT source change before and after, EVENT of SURROUND MODE and CHANNEL VOLUME does NOT return.
- E) Although EVENT of SURROUND MODE returns when the present SURROUND MODE is set up again, CHANNEL VOLUME does NOT return.
- F) When SURROUND MODE is changed, before returning SURROUND MODE after change as EVENT, the present SURROUND MODE is returned.
- G) The RESPONSE should be sent as opposed to the request command by all the commands with which an EVENT exists , not need to the another request commands(ex. SV command).
- H) The PARAMETER (with COMMAND and RESPONSE, EVENT) of minimum level of MASTER VOLUME defines "00".
- I) If the MASTER VOLUME & CHANNEL VOLUME set with 0.5dB step, the PARAMETER (with COMMAND and RESPONSE, EVENT) defines three ASCII characters as follows.

Ex. MASTER VOLUME = +18.0dB : MV98<CR>
+1.0dB : MV81<CR>
+0.5dB : MV805<CR>
0dB : MV80<CR>
-0.5dB : MV795<CR>
-1.0dB : MV79<CR>
| |
-79.5dB : MV005<CR>
--- : MV00<CR>

* At the *.0dB step, only uses two ASCII characters as PARAMETER, same as usual.

- J) 1 second later, please transmit the next COMMAND after transmitting a power on COMMAND (PWON) .

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COMMAND and RESPONSE PARAMETER list

					AVR-X4200W				AVR-X3200W				AVR-X2200W				AVR-X1200W			
COMMAND	PARAMETER	function	example	RESPONSE(example)	NA	EU	CH	JP	NA	EU	CH	NA	EU/AP	CH	JP	NA	EU/AF	CH	JP	
MAIN ZONE Control	PW	ON	POWER ON/STANDBY change	PWON<CR>	<	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	
		STANDBY		PWSTANDBY<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
	?	Return PW Status		PW?<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
	MV	UP	MASTER VOLUME UP/DOWN , direct change to **dB	MVUP<CR>	MV80<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		DOWN		MVDOWN<CR>	MV80<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		**	**00 to 98 by ASCII , 80=80(0dB), 00=0(-dB)(MIN)	MV80<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
			Default to ~Unmute (CMY)chose																	
		?	Return MV Status	MV?<CR>		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
	CV	FL UP	CHANNEL VOLUME UP/DOWN , direct change to **dB	CVFL UP<CR>	CVFL 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
			FL DOWN	--FRONT Lch	CVFL DOWN<CR>	CVFL 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
		FL **	**38 to 62 by ASCII , 50=0dB	CVFL 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FR UP	--FRONT Rch	CVFR UP<CR>	CVFR 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FR DOWN		CVFR DOWN<CR>	CVFR 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FR **	**38 to 62 by ASCII , 50=0dB	CVFR 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		C UP	--CENTERch	CVC UP<CR>	CVC 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		C DOWN		CVC DOWN<CR>	CVC 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		C **	**38 to 62 by ASCII , 50=0dB	CVC 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SW UP	--SUBWOOFER ch	CVSW UP<CR>	CVSW 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SW DOWN		CVSW DOWN<CR>	CVSW 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SW **	**00.38 to 62 by ASCII , 50=0dB.00=OFF	CVSW 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SW2 UP	--SUBWOOFER 2 ch	CVSW2 UP<CR>	CVSW2 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SW2 DOWN		CVSW2 DOWN<CR>	CVSW2 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SW2 **	**00.38 to 62 by ASCII , 50=0dB.00=OFF	CVSW2 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SL UP	--SURROUND Lch	CVSL UP<CR>	CVSL 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SL DOWN		CVSL DOWN<CR>	CVSL 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SL **	**38 to 62 by ASCII , 50=0dB	CVSL 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SR UP	--SURROUND Rch	CVSR UP<CR>	CVSR 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SR DOWN		CVSR DOWN<CR>	CVSR 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SR **	**38 to 62 by ASCII , 50=0dB	CVSR 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SBL UP	--SURROUND BACK Lch (Sbch 2SP)	CVSBL UP<CR>	CVSBL 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SBL DOWN		CVSBL DOWN<CR>	CVSBL 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SBL **	**38 to 62 by ASCII , 50=0dB	CVSBL 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SBR UP	--SURROUND BACK Rch (Sbch 2SP)	CVSBR UP<CR>	CVSBR 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SBR DOWN		CVSBR DOWN<CR>	CVSBR 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SBR **	**38 to 62 by ASCII , 50=0dB	CVSBR 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SB UP	--SURROUND BACK ch (Sbch 1SP)	CVSB UP<CR>	CVSB 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SB DOWN		CVSB DOWN<CR>	CVSB 50<CR>	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SB **	**38 to 62 by ASCII , 50=0dB	CVSB 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FHL UP	--FRONT HEIGHT Lch	CVFHL UP<CR>	CVFHL 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FHL DOWN		CVFHL DOWN<CR>	CVFHL 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FHL **	**38 to 62 by ASCII , 50=0dB	CVFHL 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FHR UP	--FRONT HEIGHT Rch	CVFHR UP<CR>	CVFHR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FHR DOWN		CVFHR DOWN<CR>	CVFHR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FHR **	**38 to 62 by ASCII , 50=0dB	CVFHR 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FWL UP	--FRONT WIDE Lch	CVFWL UP<CR>	CVFWL 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FWL DOWN		CVFWL DOWN<CR>	CVFWL 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FWL **	**38 to 62 by ASCII , 50=0dB	CVFWL 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FWR UP	--FRONT WIDE Rch	CVFWR UP<CR>	CVFWR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FWR DOWN		CVFWR DOWN<CR>	CVFWR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FWR **	**38 to 62 by ASCII , 50=0dB	CVFWR 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FTL UP	--TOP FRONT Lch	CVTFL UP<CR>	CVTFL 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FTL DOWN		CVTFL DOWN<CR>	CVTFL 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FTL **	**38 to 62 by ASCII , 50=0dB	CVTFL 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		TFR UP	--TOP FRONT Rch	CVTFR UP<CR>	CVTFR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		TFR DOWN		CVTFR DOWN<CR>	CVTFR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		TFR **	**38 to 62 by ASCII , 50=0dB	CVTFR 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		TML UP	--TOP MIDDLE Lch	CVTML UP<CR>	CVTML 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		TML DOWN		CVTML DOWN<CR>	CVTML 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		TML **	**38 to 62 by ASCII , 50=0dB	CVTML 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		TMR UP	--TOP MIDDLE Rch	CVTMR UP<CR>	CVTMR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		TMR DOWN		CVTMR DOWN<CR>	CVTMR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		TMR **	**38 to 62 by ASCII , 50=0dB	CVTMR 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		TRL UP	--TOP REAR Lch	CVTRL UP<CR>	CVTRL 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		TRL DOWN		CVTRL DOWN<CR>	CVTRL 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		TRL **	**38 to 62 by ASCII , 50=0dB	CVTRL 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		TRR UP	--TOP REAR Rch	CVTRR UP<CR>	CVTRR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		TRR DOWN		CVTRR DOWN<CR>	CVTRR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		TRR **	**38 to 62 by ASCII , 50=0dB	CVTRR 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		RHL UP	--REAR HEIGHT Lch	CVRHL UP<CR>	CVRHL 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		RHL DOWN		CVRHL DOWN<CR>	CVRHL 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		RHL **	**38 to 62 by ASCII , 50=0dB	CVRHL 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		RHR UP	--REAR HEIGHT Rch	CVRHR UP<CR>	CVRHR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		RHR DOWN		CVRHR DOWN<CR>	CVRHR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		RHR **	**38 to 62 by ASCII , 50=0dB	CVRHR 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FDL UP	--FRONT DOLBY Lch	CVFDL UP<CR>	CVFDL 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FDL DOWN		CVFDL DOWN<CR>	CVFDL 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FDL **	**38 to 62 by ASCII , 50=0dB	CVFDL 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FDR UP	--FRONT DOLBY Rch	CVFDR UP<CR>	CVFDR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FDR DOWN		CVFDR DOWN<CR>	CVFDR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		FDR **	**38 to 62 by ASCII , 50=0dB	CVFDR 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SDL UP	--SURROUND DOLBY Lch	CVSDL UP<CR>	CVSDL 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SDL DOWN		CVSDL DOWN<CR>	CVSDL 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SDL **	**38 to 62 by ASCII , 50=0dB	CVSDL 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SDR UP	--SURROUND DOLBY Rch	CVSDR UP<CR>	CVSDR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SDR DOWN		CVSDR DOWN<CR>	CVSDR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		SDR **	**38 to 62 by ASCII , 50=0dB	CVSDR 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		BDL UP	--BACK DOLBY Lch	CVBDL UP<CR>	CVBDL 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		BDL DOWN		CVBDL DOWN<CR>	CVBDL 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		BDL **	**38 to 62 by ASCII , 50=0dB	CVBDL 50<CR>	<	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		BDR UP	--BACK DOLBY Rch	CVBDR UP<CR>	CVBDR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
		BDR DOWN		CVBDR DOWN<CR>	CVBDR 50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			

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Revision
FY15V01 4/23/2015 Added X1200/X2200
FY15V02 6/30/2015 Added X3200/X4200

Volume_CMD^MV**^^Z2**^^Z3**^

Relative(dB)	Absolute
---	0
-79.5	0.5
-79	1
-78.5	1.5
-78	2
-77.5	2.5
-77	3
-76.5	3.5
-76	4
-75.5	4.5
-75	5
-74.5	5.5
-74	6
-73.5	6.5
-73	7
-72.5	7.5
-72	8
-71.5	8.5
-71	9
-70.5	9.5
-70	10
-69.5	10.5
-69	11
-68.5	11.5
-68	12
-67.5	12.5
-67	13
-66.5	13.5
-66	14
-65.5	14.5
-65	15
-64.5	15.5
-64	16
-63.5	16.5
-63	17
-62.5	17.5
-62	18
-61.5	18.5
-61	19
-60.5	19.5
-60	20
-59.5	20.5
-59	21
-58.5	21.5
-58	22
-57.5	22.5
-57	23
-56.5	23.5
-56	24
-55.5	24.5
-55	25
-54.5	25.5
-54	26
-53.5	26.5
-53	27
-52.5	27.5
-52	28
-51.5	28.5
-51	29
-50.5	29.5
-50	30
-49.5	30.5
-49	31
-48.5	31.5
-48	32
-47.5	32.5
-47	33
-46.5	33.5
-46	34
-45.5	34.5
-45	35
-44.5	35.5
-44	36
-43.5	36.5
-43	37
-42.5	37.5
-42	38
-41.5	38.5
-41	39
-40.5	39.5
-40	40
-39.5	40.5
-39	41
-38.5	41.5
-38	42
-37.5	42.5
-37	43
-36.5	43.5
-36	44
-35.5	44.5
-35	45
-34.5	45.5
-34	46
-33.5	46.5
-33	47
-32.5	47.5
-32	48
-31.5	48.5
-31	49
-30.5	49.5
-30	50
-29.5	50.5
-29	51
-28.5	51.5
-28	52
-27.5	52.5
-27	53
-26.5	53.5
-26	54
-25.5	54.5
-25	55
-24.5	55.5
-24	56
-23.5	56.5
-23	57
-22.5	57.5
-22	58
-21.5	58.5
-21	59
-20.5	59.5
-20	60
-19.5	60.5
-19	61
-18.5	61.5
-18	62
-17.5	62.5
-17	63
-16.5	63.5
-16	64
-15.5	64.5
-15	65
-14.5	65.5
-14	66
-13.5	66.5
-13	67
-12.5	67.5
-12	68
-11.5	68.5
-11	69
-10.5	69.5
-10	70
-9.5	70.5
-9	71
-8.5	71.5
-8	72
-7.5	72.5
-7	73
-6.5	73.5
-6	74
-5.5	74.5
-5	75
-4.5	75.5
-4	76
-3.5	76.5
-3	77
-2.5	77.5
-2	78
-1.5	78.5
-1	79
-0.5	79.5
0	80
0.5	80.5
1	81
1.5	81.5
2	82
2.5	82.5
3	83
3.5	83.5
4	84
4.5	84.5
5	85
5.5	85.5
6	86
6.5	86.5
7	87
7.5	87.5
8	88
8.5	88.5
9	89
9.5	89.5
10	90
10.5	90.5
11	91
11.5	91.5
12	92
12.5	92.5
13	93
13.5	93.5
14	94
14.5	94.5
15	95
15.5	95.5
16	96
16.5	96.5
17	97
17.5	97.5
18	98